

NWON Large-Scale Flexible Electrostatic Speaker for Immersive Spatial Audio

Akihisa Suzuki
Tsuchiya Co., Ltd.
Chiryu, Aichi, Japan
akihisasuzuki@tsuchiya-group.co.jp

Manabu Yoshida
National Institute of Advanced
Industrial Science and Technology
(AIST)
Tsukuba, Chiba, Japan
yoshida-manabu@aist.go.jp

Hitoshi Ito
Tsuchiya TSCO CO., Ltd.
Chiryu, Aichi, Japan
hitoshiito@tsuchiya-group.co.jp

Abstract

Conventional loudspeakers rely heavily on physical volume, weight, and placement constraints, which has hindered their seamless integration into next-generation visual and interactive environments. This paper proposes NWON, a thin, flexible electrostatic speaker that transforms fabric-like surfaces themselves into sound sources. NWON enables lightweight acoustic installations with a minimal visual footprint and easy scalability, achieving high directivity and low-distortion sound reproduction. In this work, we demonstrate the ability of large-scale NWON panels to create immersive spatial audio, proposing an integrated, space-defining audio experience at SIGGRAPH that transcends the limitations of traditional speaker placement.

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1 Introduction

Display technologies such as LCDs, LED signage, and AR/VR systems have rapidly evolved, while audio devices remain constrained by rigid structures, size, and weight. Traditional speakers require enclosures and moving diaphragms, making them visually intrusive and spatially restrictive. These limitations hinder seamless integration of sound into architectural and interactive media contexts.

NWON addresses these challenges by introducing a soft, thin, and flexible speaker that can be deployed as a surface itself. By eliminating bulky enclosures, NWON allows sound to exist naturally within space rather than as a distinct object.

2 NWON Technology

NWON is based on an electrostatic speaker principle. Its structure consists of an electret film sandwiched between conductive electrode fabrics. When an electrical signal is applied to the electrode layers, the electret film vibrates, generating sound waves.

Because the sound-generating element is a thin membrane made of flexible materials, NWON can be bent, folded, or rolled without damage. This robustness enables high durability even under rough handling. The core technology originates from research seeds developed at the National Institute of Advanced Industrial Science and Technology (AIST) in Japan, and has been further advanced in this work to achieve industrial scalability and high-fidelity acoustic performance.

3 Key Features of NWON

3.1 Surface-Integrated Sound Source

NWON does not need to be hidden within an enclosure. The surface itself becomes the sound source, allowing designers to freely choose surface materials and visual appearances according to user needs.

3.2 Low Distortion and High Directionality

Unlike conventional point-source speakers, the entire surface of NWON vibrates uniformly. This reduces attenuation and distortion, resulting in clear and precise sound reproduction. Additionally, NWON emits highly directional sound perpendicular to its surface, enabling audio to travel long distances without significant loss.

4 Scalability to Large Formats

NWON's simple layered structure allows it to be easily scaled to large dimensions. While traditional large speakers require arrays of multiple units, NWON can function as a single, continuous large-area speaker.

As speaker size increases, conventional systems often become uncomfortably loud near the source. In contrast, NWON produces sound as an aggregation of numerous small sound sources across its surface, enabling powerful output without harshness even at close distances.

A large-scale NWON installation measuring 10 m × 1 m (2-channel) was exhibited at INTER BEE IGNITION × DCEXPO 2025 at Makuhari Messe, where many visitors experienced its sound and expressed surprise and strong emotional impact.

5 Spatial Audio Using Large NWON Panels

We propose the use of NWON for spatial audio environments. Due to its flexibility and lightweight nature, NWON can be installed to surround a space. Sound waves emitted from the enclosing surfaces interfere constructively, forming an immersive surround sound field.



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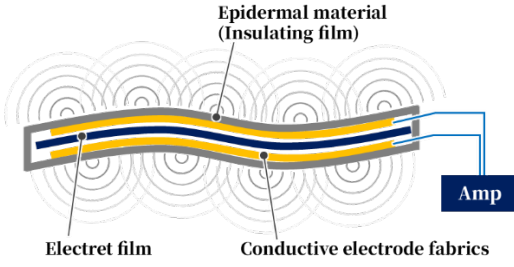
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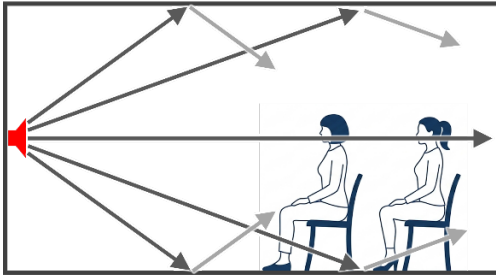
(a) Schematic cross-section of NWON.



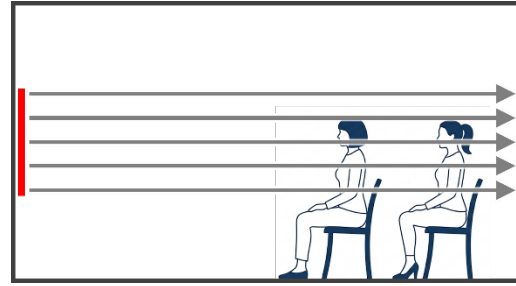
(b) Photograph of a large-scale NWON.

**Figure 1: Structural overview of NWON: (a) Schematic cross-section and (b) Photograph of a large-scale fabricated device.**

(a) Conventional cone-type speaker



(b) Proposed NWON

**Figure 2: Comparison of acoustic characteristics between a conventional cone-type speaker and the NWON. (a) Conventional cone-type speaker: Features wide sound dispersion (low directivity), significant sound attenuation, and a high tendency for reverberation. (b) NWON: Exhibits highly directional sound propagation, low sound attenuation, and minimal reverberation.**

Remarkably, such a surround experience can be achieved even with a single-channel NWON installation. As listeners move within the space, subtle changes in distance between the listener and different parts of the NWON surface create dynamic temporal shifts, resulting in the perception of moving sound sources. This unique phenomenon highlights the singularity of NWON as a speaker technology.

6 Demonstration at SIGGRAPH

At SIGGRAPH, we install a large-scale NWON panel within the booth to create a novel acoustic environment. Although the exhibition hall is filled with ambient noise, visitors experience a striking transition when stepping into the NWON space, where they are enveloped by sound and external noise appears to fade away.

This demonstration emphasizes how NWON can redefine auditory experiences in crowded public venues and interactive media exhibitions.

7 Conclusion

NWON is a groundbreaking speaker technology capable of forming immersive spatial audio using a single, flexible surface. Through its SIGGRAPH exhibition, we aim to redefine how sound functions as a next-generation interface element, seamlessly integrated into space, materials, and human experience.

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